VINCENT MULWA

Applied Physicist | Technical Systems Leader

Nairobi, Kenya | +254 707 598 xxx | mulwamutukuu@gmail.com

PROFESSIONAL PROFILE

Applied Physicist and technical systems leader with **8+ years of experience** across **battery systems, electric mobility, electronics, laboratory systems and renewable energy**. Experienced in leading technical teams, managing complex operational systems, improving reliability and translating technical performance into measurable business outcomes. Strongly adaptable across technologies, with hands-on capability in diagnostics, testing, troubleshooting, data analysis and technical problem-solving.

Education

- **MSc Physics** (Nuclear & Radiation Physics) — South Eastern Kenya University, 2023
- **BSc Physics** (Electronics Major), First Class Honours — Chuka University, 2016
- **Secondary** Education **(B+), K.C.S.E**, Top Physics Student Kenya-Kikelenzu Secondary School,2011
- **Primary** Education **(B), K.C.P.E**-Kyaitha Primary School,2007

Professional Development

- BYD Lithium Battery Training-2023
- Strathmore E-Mobility Training -2024
- AMI Management Training-2024
- Microcontroller Programming-2018
- Lean Six Sigma White Belt-2023
- Fire Marshal Training-2023

Professional Experience

Technical Lead — OhmLight Solutions | 2025–Present

Technical venture focused on battery and energy systems.

- Lead battery testing, troubleshooting, installation, maintenance and solar energy projects.
- Manage technical delivery, customer requirements, procurement and field operations.
- Built a revenue-generating operation with technicians capable of running day-to-day workshop activities independently.

Battery Manager — Electric Mobility | Ampersand Energy Kenya | 2022–2025

Led battery technical operations supporting a rapidly scaling electric-motorcycle fleet.

- Led a **12-person technical team**, setting targets, training, evaluating performance and coordinating daily delivery, and **directly supported over 25 vehicle technicians**.
- **Managed** battery testing, validation, maintenance, workshop operations and lifecycle activities **for 3,000+ batteries** supporting **2,000+ electric motorcycles** and **25+ swap stations.**
- Established a **battery health testing** and calibration capability and developed technical procedures and QA processes.
- **Reduced** average batteries out of service from **approximately 6%** to **below 3%** through **improved monitoring**, workload **visibility** and **proactive** intervention.
- Developed **real-time operational dashboards** to track battery status, workshop backlog and technician productivity, enabling earlier identification of fleet reliability risks.
- **Supported technical** decisions involving **hiring, budget, KPIs, OKRs, procurement, inventory, OPEX and CAPEX,** while reporting operational performance and risks to senior management.
- Led the implementation of a local **Battery IoT solution** leading to **cost reduction by 75%**
- **Provided technical product training** across departments and supported **engineering investigations** (battery and vehicle), **fleet reliability** and **swap-station performance**.
- **Monitored and optimized 11 solar charging plants** and used **Machine learning/ Python/Excel** for operational analysis and **customer behavior analysis**.

Technologist of Physics — South Eastern Kenya University | 2018–2022

- Managed physics/electronics laboratory operations, equipment maintenance, troubleshooting, inventory and technical support.
- Trained students in practical electronics and supported research activities.
- Established a radiation physics research unit and developed a remote-controlled digital electronics laboratory trainer.

Solar Energy Systems Intern — Plexus | 2017

- Solar Energy systems installation and Maintenance

Biomedical Engineering Attaché — Chuka Referral Hospital | 2015

- Biomedical engineering systems and equipment installation and maintenance

Professional Engagement

- Technical Committee Member — KEBS TC199 (Electric Mobility)
- Technical Committee Member — KEBS TC091 (Electrical Installations and Distribution Systems)
- Member — Electric Mobility Association of Kenya (EMAK)
- Technical Lead-Kenya Electric Mobility Technical Group (KEMTG)

Expertise

- Technical Leadership, Team Management, and team Capacity Building
- Scientific Research (Experiment design, execution and reporting)
- Battery Lifecycle Management for sustainable transport and energy storage systems
- Data-Driven Strategies for continuous product improvement.
- Productivity Management & Operations Optimization

Recent Technical Execution

- June-2026 **Testing and commissioning** of Electric three wheelers for a farm operations-Machakos Region
- April-2026 **Rivian Fleet Service** Technical Support-Nairobi and Chyulu Region
- Mar-2026 **Sensor and DAQ Instrumentation on EV and ICE** Motorcycles for Road and Load Data Acquisition (**RLDA**) Test Nairobi, Coast and Rift Valley Region
- Jan 2026 **DAQ Instrumentation on EV and ICE** for Real User Profile Analysis (RUPA) Test- Nairobi Region
- Nov-2025 **DAQ Instrumentation on ICE** for Real User Profile Analysis (**RUPA**) Test Eastern Region

Hobbies

- High Speed Driving
- Tinkering with electronics devices, batteries, Algorithms and automation

References

- Available upon Request